

CAPE POINT, South Africa

WMO: 689160

Lat: 34.3533S

Lon: 18.4894E

Elev: 748

StdP: 14.30

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	47.3	48.7	32.0	27.1	64.3	35.6	31.4	59.7	47.9	55.2	44.5	53.7	16.3	230	0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.3	80.5	63.5	76.9	63.4	74.3	63.5	68.6	73.7	67.5	72.3	66.4	71.0	12.8	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.2	102.7	70.4	65.8	97.8	69.4	64.8	94.3	68.8	33.3	74.5	32.4	72.4	31.5	71.2	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 51.7	46.8	41.3	DB	44.8	89.5	1.7	4.4	43.6	92.7	42.6	95.2	41.7	97.7	40.4	100.8	(4)
(5)			WB	41.0	71.1	1.6	1.3	39.9	72.0	39.0	72.8	38.1	73.5	37.0	74.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	61.0	66.8	67.0	65.2	62.6	59.8	56.8	55.8	55.5	57.1	59.6	61.6	65.2	(6)
(7)		DBStd	5.44	2.74	2.85	3.67	4.60	3.64	3.57	4.07	3.62	3.92	3.63	3.18	3.25	(7)
(8)		HDD50	5	0	0	0	0	0	1	2	2	0	0	0	0	(8)
(9)		HDD65	1727	11	10	39	100	170	250	286	296	244	176	109	36	(9)
(10)		CDD50	4033	521	475	471	377	303	204	183	172	212	296	348	471	(10)
(11)		CDD65	281	67	65	45	28	8	3	2	1	6	7	7	42	(11)
(12)		CDH74	696	92	95	163	147	26	7	3	5	46	36	22	54	(12)
(13)		CDH80	149	18	13	51	38	1	0	0	0	13	4	4	7	(13)
(14)	Wind	WSAvg	17.4	16.8	19.5	18.9	18.7	16.8	17.3	17.0	17.1	15.6	17.3	17.5	16.9	(14)
(15)	Precipitation	PrecAvg	26.4	0.8	0.6	0.9	2.0	3.5	4.0	4.4	4.3	2.5	1.3	1.3	0.8	(15)
(16)		PrecMax	35.2	3.8	1.4	2.9	5.1	6.6	7.2	9.4	8.0	5.9	5.0	3.9	1.7	(16)
(17)		PrecMin	17.4	0.0	0.0	0.0	0.5	0.2	1.1	1.6	2.0	0.8	0.0	0.1	0.0	(17)
(18)		PrecStd	4.8	0.9	0.4	0.7	1.1	1.4	1.3	1.9	1.8	1.3	1.0	1.0	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.3	81.9	86.3	84.7	78.1	74.2	73.3	73.5	80.4	79.2	76.4	79.7	(19)
(20)			MCWB	65.6	65.9	66.2	62.2	60.2	56.7	54.6	56.2	58.2	61.4	62.4	67.0	(20)
(21)		2%	DB	76.2	76.4	78.5	77.8	72.9	69.3	69.0	67.9	70.2	72.4	71.0	75.0	(21)
(22)			MCWB	66.8	65.7	64.7	61.0	57.2	53.8	53.7	52.9	55.9	60.0	62.0	66.1	(22)
(23)		5%	DB	73.2	73.7	73.3	72.9	69.0	65.2	65.4	63.2	64.6	67.5	68.3	72.4	(23)
(24)			MCWB	65.9	65.5	64.6	60.6	56.7	52.7	52.8	53.9	56.6	59.5	61.3	64.8	(24)
(25)		10%	DB	71.4	71.6	70.4	68.8	65.3	61.6	61.6	60.3	61.6	64.7	66.5	70.4	(25)
(26)			MCWB	65.0	64.7	63.5	60.8	57.1	54.5	53.7	54.4	55.8	58.8	60.4	63.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.7	69.3	69.1	66.1	62.9	59.9	59.0	59.1	61.7	64.5	65.8	69.5	(27)
(28)			MCDWB	75.1	74.4	76.6	74.5	69.8	65.4	63.1	64.1	71.3	72.7	72.2	75.9	(28)
(29)		2%	WB	68.8	68.0	67.3	64.3	61.1	58.3	57.3	57.6	59.3	62.3	63.7	67.5	(29)
(30)			MCDWB	72.8	72.5	73.3	71.0	65.7	62.2	61.3	61.5	65.4	67.9	68.7	73.0	(30)
(31)		5%	WB	67.5	67.0	65.9	63.1	60.0	57.2	56.2	56.5	57.9	60.8	62.5	66.1	(31)
(32)			MCDWB	71.4	71.5	70.8	68.8	63.9	60.5	60.0	60.1	62.8	65.7	66.9	70.8	(32)
(33)		10%	WB	66.2	65.9	64.8	61.9	58.9	56.2	55.2	55.3	56.6	59.5	61.2	64.9	(33)
(34)			MCDWB	70.0	70.4	69.4	67.0	62.6	59.5	59.0	58.8	60.8	63.9	65.4	69.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.3	9.9	9.7	9.7	8.3	7.5	7.8	8.1	8.5	9.1	9.1	9.6	(35)
(36)			MCDBR	12.1	13.6	15.0	17.2	14.8	13.9	14.3	13.6	14.3	14.1	12.3	12.3	(36)
(37)			MCWBR	6.1	6.5	6.7	7.0	6.1	6.8	6.4	7.0	6.9	6.8	6.9	6.5	(37)
(38)		5% WB	MCDWR	10.4	11.6	12.1	12.8	9.9	8.7	9.1	9.8	11.7	12.2	11.3	11.5	(38)
(39)			MCWBR	6.0	6.5	6.4	6.6	6.2	6.2	6.0	6.3	6.7	6.7	7.0	6.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.336	0.338	0.328	0.317	0.316	0.310	0.312	0.325	0.335	0.344	0.333	0.337	(40)	
(41)		taud	2.580	2.581	2.606	2.630	2.618	2.605	2.599	2.562	2.498	2.488	2.564	2.566	(41)	
(42)		Ebn at Noon	318	310	302	288	271	263	269	281	295	305	317	318	(42)	
(43)		Edh at Noon	33	32	29	25	23	22	23	27	32	35	34	34	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2557	2271	1806	1321	918	767	855	1097	1521	2008	2387	2594	(44)	
(45)		RadStd	55	69	78	62	55	46	48	54	63	90	76	58	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		+0.65	+0.88	N/A	N/A	+0.95	+0.85	-4	-133	+200	+110

Nomenclature: See separate page